Abhishek Kumar

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Research Interests

Computational Neuroscience, Neuromorphic Computing, Spiking Neural Network, Neural Modeling, Machine Learning, Brain-Inspired Artificial Intelligence

Education

ABV-Indian Institute of Information Technology and Management Gwalior — Gwalior, MP
MS (Artificial Intelligence and Data Science) — *Aug. 2025 - Present*

University of Delhi — Delhi
Bachelor of Science, Mathematics (H) — *Nov. 2022 - Jun 2025*

Experience/Projects

Winter Research Internship Certificate — Dec. 2025 - Jan. 2026
Ion channel and Receptor biology lab, KSBS — *Indian Institute of Technology, Delhi*

- Developed neuron simulations using **NEURON** to study ion-channel and spike dynamics.
- Implemented and modified biophysical neuron models using **hoc scripting** and explored ModelDB.
- Applied supervised learning models to molecular datasets for toxicity prediction, aligning with ML-driven drug discovery.

Aadiyogi — Dec. 2022 - Jun. 2025
Team Captain — *Deshbandhu College, UoD*

- Led the team in regional, state, and national-level yoga competitions and championships.
- Facilitated weekly practice sessions to discuss and master advanced Asanas, Pranayama & Meditation techniques.
- Acted as the primary liaison between team members, coaching staff, and competition officials.

Projects

Ultra-Low-Power Edge Intelligence for Real-World Anomaly Detection Using SNN — Ongoing
Spiking Neural Networks, Edge AI, Embedded Systems Project info.

- Creating an event-driven anomaly detection framework to identify irregular patterns that indicate sewage blockages.
- Project currently under development; implementation details and results will be shared upon completion.

Smart Shield: Lightweight On-Device SMS Fraud Detection Using SNN — Mar. 2026
GitHub — *Hacksecure'26 Hackathon, NIT-H*

- Developing an Android application to detect fraudulent and spam SMS messages.
- Integrated a snnTorch-based classification model for real-time message analysis.
- Implemented preprocessing and inference pipeline for on-device ML deployment.
- Built during Hacksecure'26, a cybersecurity hackathon organized under ISEA Phase-III (MeitY, Govt. of India).

Multi-Objective Analysis of Spiking Neuron Models (Python) — Jan. 2026
GitHub — *Research Project*

- Built multi-objective evaluation framework for SNN models.
- Compared models on efficiency and biological realism.
- Simulated responses under varied stimulation protocols.
- Analyzed spike trains and membrane dynamics.

Spiking Neuron Simulations (Python, Brian2 / NEURON) Dec. 2025 - Present
GitHub *Self-Directed Practice Repository*

- Maintained a practice repository containing multiple implementations of spiking neuron models under varied conditions.
- Explored neuron behavior across different parameter regimes, input currents, and initial conditions.
- Used visualization of membrane potentials and spike trains to analyze model behavior.

Early-Stage Heart Attack Prediction (Python, ML) Oct. 2025
GitHub *Machine Learning Project*

- Built classification models for early heart attack prediction.
- Performed data preprocessing and feature engineering.
- Implemented CatBoost and XGBoost for improved accuracy.
- Optimized models using attention-based enhancements.

Publications

Abhishek Kumar, "Computational Neuroscience–Inspired Neuromorphic Computing: A Review of Models, Hardware Architectures, and Applications". **(Manuscript submitted to a journal; Presented at ICMSCI 2026)**

Conference and Academic Events

ICMSCI'26 Certificate Feb. 2026
Paper Presented *PGDAV College, UoD*

* Presented a paper at International Conference on Mathematical Sciences and Computational Intelligence.
* Topic: Computational Neuroscience–Inspired Neuromorphic Computing.
* Analyzed TrueNorth, Loihi, SpiNNaker, and Neurogrid architectures.
* Focused on energy-efficient AI and edge applications.

MATHSD'25 Certificate Oct. 2025
Volunteering *ABV-IIITM Gwalior*

* Volunteered at MATHSD 2025, hosted by ABV-IIITM Gwalior.
* Handled LaTeX typesetting and compilation of the official **Book of Abstracts**.
* Assisted faculty and research teams in documentation and on-site coordination.
* Gained insights on *Neuroscience-Inspired Scientific Machine Learning* through keynote sessions.

Samuchchayam 2025 Certificate March 2025
Presented Paper *Deshbandhu college, UoD*

* Awarded first prize in the **'Datamatrix'** paper presentation competition.
* Recognized for excellence in research, analysis, and public speaking at a competitive academic event
* Successfully developed and presented a compelling paper on a data-centric topic to a panel of faculty and peers.

InitMath 2024 Certificate March 2024
Mathematics Camp *Uttarkashi (U.K)*

* Selected for and completed the 'Initiation into Mathematics (InitMath) 2024' training program.
* Participated in an intensive workshop sponsored by the prestigious **National Board for Higher Mathematics (NBHM)**.
* Engaged in a 6-day program focused on foundational concepts in advanced mathematics, organized by **MTTS Trust**.

Training and Certifications

Computational Neuroscience – University of Washington Certificate Nov. 2025 - Jan. 2026

- Strengthened understanding of the intersection between Neural biology and computation.
- Instructors: Rajesh P. N. Rao, Adrienne Fairhall

Introduction to Cognitive Psychology and Neuropsychology - University of Cambridge Jan. 2026 - Present

- Learned theories, models and findings of cognitive functions, including memory, attention and perception.
- Instructors: Dr. Giulia Mangiaracina

Linear Algebra for Machine Learning – LDRP Institute of Technology & Research Certificate Dec. 2025

- National Short-Term Training Program
- Hands-on experience on PCA, SVD, and image processing.

Achievements & Awards

National Yoga Championship Runner up Certificate Sept. 2023

- Championship held in Haridwar and organised by **World Fitness Federation of Yogasana Sports, India**

Certificate of Commendation by Chief Minister of Uttarakhand Certificate June 2020

- Award presented to students with outstanding achievements in academics.

Skills & Interests

Languages: Python, MySQL, R

Tools: Power BI, Excel, Mathematica, MATLAB

Frameworks & Libraries: Numpy, Pandas, Matplotlib, Seaborn, Scikitlearn, PyTorch, snnTorch, NEURON

Domains: Computational Neuroscience, Machine Learning, Deep Learning, Neural Network

Others: Kaggle, Apache Kafka , Git, GitHub